8/9/2023

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CHEMISTRY

Section - A

Q1.) (4) Argon

Q2.) (2) Calcium

Q3.) $CO_2 = 12 + 32$
$= 44g$

$VD = \frac{mass}{2} = 22$

(4) 22

Q4.) 19 - 2,8,8,1
(1) 17

Q5) (3) 1:1

Q6) (4) Mercury

Q7) (2) Lead

Q8) (1) 1

Q9.) (1) Lithium

Q10.) (4) Ammonium chloride

Q11.) (4) Copper chloride

①

②

Q12) (2) Consists of molecules

Q13) (3) An alkali

Q14) (1) Direct combination

Q15) (3) conc. HCl + conc. $HNO_3$

Q16) (i) a) Z

b) L

c) Halogens

d)

| | group 13 | group 15 |
|---|---|---|
| Period 2 | Boron | ~~Fluorin~~ Nitrogen |
| Period 3 | Aluminium | Phosphorus |
| ~~Period 4~~ | | |
| ~~Period 5~~ | | |

e) O

(ii) a) reducing agents

b) bad

c) acidic

d) Silver chloride

e) Cl

(iii) $SO_2 = 32 + 32$
$= 64g$

| Volume | g |
|---|---|
| 22.4 | 64 |
| x | 320 |

$$x = \frac{22.4 \times \cancel{320}^{5}}{\cancel{64}_{8}}$$

$\therefore$ volume = 112 l

(ii) When gases react (combine), they do so in volume, which bear a simple whole number ratio to one another and also with the gaseous product, provided that their volumes are measured ~~in~~ at same temperature and pressure

(iii) $C_3H_8 + 5O_2 \longrightarrow 3CO_2 + 4H_2O$

$C_3H_8 = 36 + 8 = 44\,g$

| $C_3H_8$ | $O_2$ |
|---|---|
| 44 g | $5 \times 22.4$ |
| 8.8 g | $x$ |

$$x = \frac{\cancel{8.8}^{0.2} \times 5 \times 22.4}{\cancel{44}}$$

$= 0.2 \times 112$

$\therefore$ Volume of $O_2$ = 22.4 l

iv) (i) Na = 23 g

| g | atoms |
|---|---|
| 23 | $6.022 \times 10^{23}$ |
| 4.6 | $x$ |

$$x = \frac{\cancel{4.6}^{0.2} \times 6.022 \times 10^{23}}{\cancel{23}}$$

$= 2.4088 \times 10^{23} = 24.088 \times 10^{22}$

$\therefore$ Ans = $24.088 \times 10^{22}$

(i) $CuSO_4$
$= 64 +$
$= 96 +$
$= 160$
$= 250$

% of

(ii) Emp
VD =

molecula

$\therefore$ molec

v) (i) 15
(ii) 19
(iii) 8
(iv) 4
(v) 2

Section

1) molecu
Emp

Emp we

(4)

(ii) $CuSO_4 \cdot 5H_2O$

$= 64 + 32 + 64 + 5(18)$

$= 96 + 64 + 90$

$= 160 + 90$

$= 250$

% of water of crystallization $= \frac{\cancel{90}^{18}}{\cancel{250}_{5}} \times \cancel{100}^{2}$

$= 36\%$

(iii) emp formul = $XY_2$

VD = emp weight

molecular weight = $2 \times VD$

$= 2 \times$ emp weight.

$\therefore$ molecular formula $= 2 \times XY_2$

$= X_2Y_4$

v) (i) 15

(ii) 19

(iii) 8

(iv) 4

(v) 2

Section - B

1) molecular wgt = 108

Emp formula = $C_3H_4N$

emp weigth = $36 + 4 + 14 = 54g$

volume, which

e another and

hat their volumes

and pressure

$H_2O$

2 2

5

molecular weight = 2 × VD

VD = 54

$n = \cancel{\frac{\text{mole emp weight}}{\text{molecular}}}$

$n = \frac{\text{molecular weight}}{\text{emp. weight}}$

$= \frac{108}{54}$ 2

∴ n = 2

Molecular formula = $2 \times C_3H_4N$

$= C_6H_8N_2$

in 1 mole of $C_6H_8N_2$, amount of carbon = 6×12

= 72 g

ii) a) 1. pH will decrease

2. pH will increase

b) group 15

iii) $2Ca(NO_3)_2 \longrightarrow 2CaO + 4NO_2 + O_2$

8.2 g

$Ca(NO_3)_2 = 40 + 28 + 96$

$= 164$ g

$NO_2 = 14 + 32$

$= 46$ g

$n = 6 \times 12$

$= 72g$

$Ca(NO_3)_2$ $\quad$ $NO_2$

$2 \times 164$ $\quad$ $4 \times 22.4$

$8.2$ $\quad$ $x$

$$x = \frac{\cancel{8.2}^{0.1} \times \cancel{4}^{2} \times 22.4}{\cancel{2} \times \cancel{164}_{82}}$$

(a) Volume $= 2.24 \ l$

b) $Ca(NO_3)_2$ $\quad$ $CaO$

$2 \times 164$ $\quad$ $2 \times 56$

$8.2$ $\quad$ $x$

$$x = \frac{\cancel{8.2}^{0.1} \times \cancel{2} \times 56}{\cancel{2} \times \cancel{164}_{82}}$$

$= 0.05 \times 56$

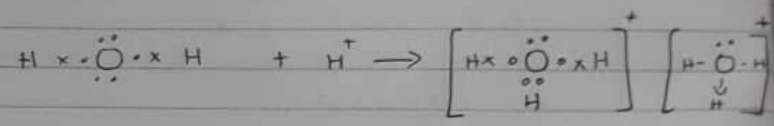
$\therefore$ Mass $= 2.8g$

iv) i) a) No, it can not be differentiated because both are amphoteric in nature. They can react with acid and base.

b) yes.
When ~~Ca~~ Calcium nitrate reacts with ammonium hydroxide solution, pale blue ppt. is formed. and in excess inky blue soln is formed.
when lead nitrate reacts with ammonium hydroxide solution, chalky white ppt. is formed and in excess it is insoluble.

ii)

ii) $H_2O + H^+ \longrightarrow H_3O^+$



Q2.) i) 1. ~~True~~ False

~~Noble gas have complete shells and are stable~~

$Na (2,8,1) \longrightarrow Na^{1+} (2,8)$ The Noble gas in that period has (2,8,8,2)

2. ~~Ele~~ False

~~They are to~~ it is a covalent compound and hence a bad/ poor conductor of electricity.

ii)

| | |
|---|---|
| g | moles |
| 44 | 1 |
| 22 | $x$ |

$x = \frac{22 \times 1}{44} = \frac{1}{2} = 0.5$

$\therefore$ 0.5 moles in 22g of $CO_2$

iii) a) As element with high electron affinity readily accepts electron, so it has higher oxidising power.

$X > Y$

b) As more electron affinity element shows more ~~elec~~ electronegetivity.

$X > Y$

c) X accepts electrons easily hence X is non-metal. In P.T. left side elements are ~~non~~-metals and right side elements are non-metals.

Hence, X is right of Y in P.T.

Q3.) i) $O^{2-}$

NO =>

$M^{2+}$ +

$M^{2+}$ +

iii) (i) $MgCl_2$

Mg +

(ii) CaO

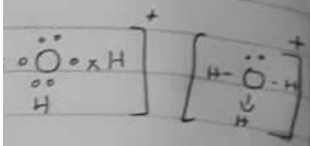
Ca

(iii) NaC

Na

iv) A type are antis

Eg : N



and are stable
) The Noble gas in that
(2,8,8,2)

and hence a bond/

ly readily accepts
power

shows more

is non-metal
metals and right



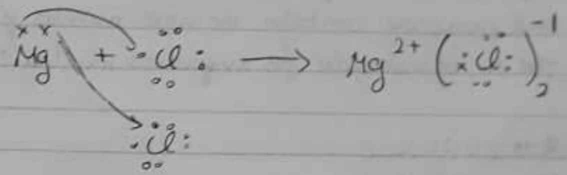
Q3.) i) $O^{2-}$

$MO \Rightarrow O^{2-} + M^{2+} \longrightarrow MO$

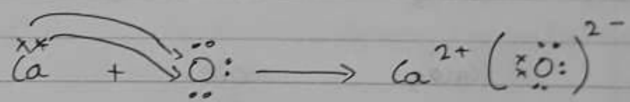
$M^{2+} + Cl_2^{1-} \longrightarrow MCl_2$

$M^{2+} + S^{2-} \longrightarrow MS$

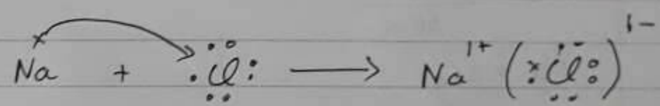
ii) (i) $MgCl_2$



(ii) CaO



(iii) NaCl



iv) A type of covalency in which the shared pair of electrons are entirely provided by one of the combining atoms.
Eg : $NH_4^+$, $OH^-$, $H_3O^+$

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Q4.) i)(a) 2, 8, 8, 2
(b) period - 4
group - 2

ii)(a) 2
(b) ionic bond

iii) Equal volume of all gases under the same temp and pressure contain the same number of molecules.
The no. of molecules for avogadros number is $6.022 \times 10^{23}$.

iv.) ~~(i) B, D~~ (i) A, C
~~(ii)~~ (ii) B, D
(iii) B, D

Q5.) i) $CaCO_3 \longrightarrow CaO + CO_2$

| | |
|---|---|
| $CaCO_3$ | $CaO$ |
| 100g | 56g |
| $x$ | 14g |

$$x = \frac{\cancel{100}^{25} \times \cancel{14}}{\cancel{56}_{\cancel{4}}}$$

$\therefore x = 25g$

ii) $O_2 + C \longrightarrow CO_2$
20g 20g 20g

$CO_2$ produced is 20g

iii) $Na_2SO_3$

$Na_2$

$x$

$x =$

$\therefore$ mass of $Na_2SO_3$ = 5.8

iv)(i) Copper
(ii) normal
acid sa
(iii) Lead

Q6.) i) (a) Copper
(b) Sodium

ii) Deliques
directly

iii) *. Turn
*. Sour
*. Turns

iv)

(9)

...nder the same temp
...umber of molecules.
...umber is $6.022 \times 10^{23}$.

(10)

iii) $Na_2SO_3 + H_2SO_4 \longrightarrow Na_2SO_4 + H_2O + SO_2$

| $Na_2SO_3$ | $SO_2$ |
|---|---|
| 116g | 64g |
| $x$ | 3.2g |

$$x = \frac{116 \times 3.2}{64}$$

$\therefore$ mass of $Na_2SO_3$ = 5.8g

iv) (i) Copper hydroxide, aluminium hydroxide
(ii) normal salt $Na_2CO_3$ - Sodium Carbonate
acid salt $NaHCO_3$ - Sodium bicarbonate.
(iii) Lead chloride

Q6.) i) (a) Copper ~~Sulphate~~ (II) Sulphate Crystals - $CuSO_4.5H_2O$
(b) Sodium bicarbonate

ii) Deliquescence is the process in which a solid substance directly convert into gaseous state.

iii) *. Turn blue litmus red.
*. Sour in taste
*. Turns myrthyle orange red

iv)